Master's Thesis Progress Documentation

Phase 5: Explainable Artificial Intelligence with SHAP

1. Thesis Title

Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence

2. Purpose of This Document

This document summarizes the work completed during Phase 5 of the master's thesis.

The main purpose of this phase was to explain the customer churn predictions produced by the selected Gradient Boosting model. During Phase 4, the machine learning model generated customer-level churn probabilities and predicted churn labels. However, those predictions alone did not explain why an individual customer received a high or low churn probability.

In Phase 5, SHAP was used to calculate the contribution of each input feature to the model prediction. These contributions show which customer characteristics increased the predicted churn risk and which characteristics reduced it.

The explanations were created at two levels:

- Processed model-feature level
- Grouped original-feature level

The grouped explanations were connected to customer IDs, original customer values, actual churn status, predicted churn status, and churn probability.

Power BI-ready output files were also created. These files will be used in the next phase to develop the interactive visual analytics dashboard.

3. Current Stage of the Thesis

The thesis has now completed the following stages:

- Research foundation and planning
- Dataset selection
- Exploratory Data Analysis
- Data cleaning
- Data preprocessing
- Machine learning model development
- Explainable Artificial Intelligence analysis using SHAP

Phase 5 created the explainability component of the proposed visual analytics system.

The current project pipeline is:

Customer Data
→ Data Preprocessing
→ Gradient Boosting Model
→ Churn Predictions and Probabilities
→ SHAP Feature Contributions
→ Structured Explanation Outputs
→ Power BI Dashboard
→ User Evaluation

The selected Gradient Boosting model remains responsible for generating churn probabilities.

SHAP is responsible for explaining how the customer features influenced those probabilities.

Power BI will be responsible for displaying the predictions and explanations through an interactive user interface.

4. Work Completed in Phase 5

4.1 Creation of the SHAP Analysis Notebook

A new Google Colab notebook was created for the Explainable Artificial Intelligence phase.

The notebook was named:

04_shap_analysis.ipynb

The notebook contains 19 completed code cells.

All SHAP preparation, calculation, validation, visualization, customer-level explanation creation, output generation, and final file validation were completed inside this notebook.

The notebook was designed to run using the saved outputs from Phase 3 and Phase 4.

4.2 Environment and Library Setup

The Google Colab environment was checked before starting the analysis.

The main environment versions were:

- Python: 3.12.13
- Pandas: 2.2.2
- NumPy: 2.0.2
- Scikit-learn: 1.6.1
- SHAP: 0.52.0

- Joblib: 1.5.3

A random state of 42 was used to support reproducibility.

The SHAP library was already installed in the environment and did not require an additional installation.

4.3 Loading the Phase 3 and Phase 4 Outputs

The complete output ZIP archives from the previous phases were uploaded and extracted.

The uploaded archives were:

- Phase_3_Preprocessing_Outputs.zip
- Phase_4_Model_Development_Outputs.zip

The Phase 3 archive contained 10 files, and the Phase 4 archive contained 15 files.

All required files were detected successfully.

The main Phase 3 files used during this phase were:

- X_train_processed.csv
- X_test_processed.csv
- y_train.csv
- y_test.csv
- train_reference.csv
- test_reference.csv
- feature_name_mapping.csv
- preprocessing_metadata.json

The main Phase 4 files used were:

- selected_model.joblib
- model_feature_names.csv
- test_predictions.csv
- test_customer_predictions.csv
- training_oof_predictions.csv
- best_model_parameters.json
- model_metadata.json

4.4 Loading the Selected Model and Datasets

The final trained model selected during Phase 4 was loaded successfully.

The model class was:

GradientBoostingClassifier

The model explained the positive class:

Churn = 1

The selected prediction threshold remained:

0.27

The loaded datasets contained:

Dataset	Records	Features
Processed training data	5,625	45
Processed testing data	1,407	45
Training reference data	5,625	22
Testing reference data	1,407	22

The testing prediction file contained 1,407 customer records.

The customer prediction file contained the original customer information together with the model probability and predicted churn result.

4.5 Model and Data Validation

Before calculating SHAP values, several validation checks were completed.

The checks confirmed that:

- The selected Gradient Boosting model loaded successfully.
- The training dataset contained 5,625 rows and 45 features.
- The testing dataset contained 1,407 rows and 45 features.
- The training and testing targets matched their corresponding feature datasets.
- The training and testing feature columns were in the correct order.
- The 45 feature names matched the saved Phase 4 model feature order.
- The customer IDs remained aligned.
- The actual churn values remained aligned.
- The processed data contained no missing values.
- The processed data contained no infinite values.
- The positive churn class was correctly identified as class 1.

The probabilities generated by the loaded model were compared with the saved probabilities from Phase 4.

The maximum probability difference was approximately:

9.71×10^{-17}

This value is effectively zero and confirms that the exact Phase 4 model was loaded correctly.

The predicted labels created using the threshold of 0.27 also matched the saved Phase 4 results.

4.6 Creation of the SHAP Feature Mapping

The model used 45 processed numerical features.

These features included:

- Three standardized continuous variables
- One binary numerical variable
- One-hot encoded categorical variables

The three continuous features were:

- Tenure
- Monthly Charges
- Total Charges

The binary numerical feature was:

- Senior Citizen

The remaining processed features were created through one-hot encoding.

For example, the original feature Contract was represented through processed columns such as:

- Contract_Month_to_month
- Contract_One_year
- Contract_Two_year

A complete feature-mapping table was created to connect every processed feature to its original customer feature.

The mapping contained:

- Processed feature position
- Processed feature name
- Original feature name
- Readable feature display name
- Encoded category
- Feature type

All 45 processed features were successfully connected to the 19 original input features.

4.7 Selection of the SHAP Explainer

The selected model was tree-based. Therefore, SHAP TreeExplainer was used.

The final explainer configuration was:

- Explainer: TreeExplainer
- Feature perturbation method: Interventional
- Model output: Probability
- Background sample: 100 training records
- Random state: 42
- Positive class: Churn = 1

A reproducible sample of 100 training customers was used as the SHAP background dataset.

The interventional approach was selected because it allowed the SHAP values to be calculated directly on the churn-probability scale.

4.8 SHAP Output Scale

The SHAP values were successfully calculated on the:

Probability scale

This means that each SHAP contribution can be interpreted as an increase or decrease in the model's predicted churn probability relative to the SHAP base value.

The SHAP base value was approximately:

0.3203

For an individual customer, the explanation follows this structure:

Base probability + sum of SHAP feature contributions = customer churn probability

A positive SHAP value means that the feature increased the predicted churn probability.

A negative SHAP value means that the feature reduced the predicted churn probability.

For example:

- SHAP value of +0.08 means that the feature increased the churn probability by approximately 0.08, or eight percentage points.
- SHAP value of −0.05 means that the feature reduced the churn probability by approximately 0.05, or five percentage points.

The prediction threshold of 0.27 was not used to calculate the SHAP values.

The threshold was used only to convert the final probability into a churn or non-churn prediction.

4.9 SHAP Value Calculation

SHAP values were calculated for the complete independent test dataset.

The result contained:

- 1,407 customer explanations
- 45 processed feature contributions for each customer
- 63,315 total processed-feature SHAP values

The final SHAP array had the shape:

$1{,}407 \times 45$

The output was checked carefully because different SHAP versions may return different data structures.

The notebook safely handled the SHAP Explanation object and created a consistent two-dimensional customer-by-feature structure.

The calculation confirmed that:

- Every test customer received an explanation.
- Every customer received one SHAP value for each of the 45 features.
- The explanations represented the positive churn class.
- No SHAP values were missing.
- No infinite SHAP values were present.

4.10 SHAP Additivity Validation

SHAP additivity was validated for all 1,407 test customers.

The following relationship was checked:

$$\text{Base value} + \text{sum of SHAP contributions} \approx \text{model churn probability}$$

The additivity tolerance was:

0.000001

The results were:

- Customers validated: 1,407
- Customers passing validation: 1,407
- Customers failing validation: 0
- Maximum additivity error: approximately 3.38×10^{-9}
- Mean additivity error: approximately 9.08×10^{-10}

The maximum error was much smaller than the accepted tolerance.

This confirmed that the exported SHAP values correctly reconstructed the model's churn probability for every customer.

A separate additivity-validation file was created containing:

- Customer ID
- SHAP output scale
- Base value
- Sum of SHAP contributions
- Reconstructed model output
- Original model output
- Absolute additivity error
- Validation result

4.11 Global Processed-Feature Importance

Global feature importance was first calculated using the 45 processed model features.

The importance measure used was:

Mean absolute SHAP value

Mean absolute SHAP measures the average strength of each feature's contribution across all test customers.

The strongest processed model features were:

Rank	Processed feature
1	tenure
2	Contract_Month_to_month
3	InternetService_Fiber_optic
4	OnlineSecurity_No
5	TechSupport_No
6	PaymentMethod_Electronic_check
7	MonthlyCharges
8	Contract_Two_year
9	MultipleLines_No
10	TotalCharges

This technical feature representation preserves the exact columns used by the machine learning model.

However, separate one-hot encoded columns can be difficult for non-technical users to understand. Therefore, a second grouped explanation level was also created.

4.12 Grouping SHAP Values into Original Features

The 45 processed SHAP features were grouped into the 19 original customer features.

For categorical features, the contributions of the related one-hot encoded columns were added together.

For example, the following processed contributions:

- Contract_Month_to_month
- Contract_One_year
- Contract_Two_year

were grouped into:

Contract

The grouping process preserved the total SHAP contribution for every customer.

The maximum difference between the processed-feature total and the grouped-feature total was approximately:

$$2.22 \times 10^{-16}$$

This is effectively zero and confirms that the grouped explanations remained mathematically consistent with the original SHAP explanations.

The grouped feature level is more suitable for Power BI because users can view a single readable feature such as Contract instead of several technical one-hot encoded columns.

4.13 Global Grouped Feature Importance

Global importance was calculated for the 19 grouped original features.

The main results were:

Rank	Original feature	Mean absolute SHAP
1	Tenure	0.086874
2	Contract	0.083423
3	Internet Service	0.048631
4	Online Security	0.035766
5	Technical Support	0.029579
6	Payment Method	0.025229
7	Monthly Charges	0.024565
8	Multiple Lines	0.011716
9	Paperless Billing	0.010842
10	Total Charges	0.007523

The results show that Tenure and Contract were the two most influential original features in the final model.

Internet Service, Online Security, Technical Support, Payment Method, and Monthly Charges also made important contributions to the churn predictions.

The following features received a mean absolute SHAP value of zero in the fitted model:

- Gender
- Partner
- Device Protection

This means that these variables did not contribute to the final predictions generated by this specific trained model for the test customers.

This result should not be interpreted as a general statement that these features can never influence churn. It only describes their role in the selected model and dataset.

4.14 Interpretation of the Global Results

The global SHAP results were consistent with the business patterns observed during the exploratory analysis and machine learning phases.

The strongest factors were related to:

- Customer relationship duration
- Contract commitment
- Internet service
- Availability of online security
- Availability of technical support
- Payment method
- Monthly customer charges

Low customer tenure often contributed to higher churn risk.

Longer tenure generally contributed to lower churn risk.

Month-to-month contracts were among the strongest processed model features.

This indicates that customers without a longer-term contract frequently received a higher churn-risk contribution.

Internet service type also played an important role. Fibre-optic service was one of the strongest processed features.

The absence of online security and technical support also influenced the model's churn predictions.

Electronic-check payment was another important processed feature.

The results indicate which variables the model relied on most strongly. They do not prove that these features directly cause customer churn.

4.15 Global SHAP Visualizations

Three main global SHAP visualizations were created.

Processed-Feature Global Bar Plot

This plot displays the strongest processed model features according to mean absolute SHAP value.

It preserves the technical feature structure used by the model.

SHAP Beeswarm Plot

The SH’s churn predictions.

Electronic-check payment was another important processed feature.

The results indicate which variables the model relied on most strongly. They do not prove that these features directly cause customer churn.

4.15 Global SHAP Visualizations

Three main global SHAP visualizations were created.

Processed-Feature Global Bar Plot

This plot displays the strongest processed model features according to mean absolute SHAP value.

It preserves the technical feature structure used by the modelAP beeswarm plot displays:

- Global feature importance
- Distribution of feature contributions
- Positive and negative contribution direction
- Relative customer feature values

This visualization helps show both how important a feature is and how its values influence different customers.

Grouped Original-Feature Bar Plot

This plot displays global importance using the 19 readable original features.

It is easier to understand and will be more suitable for the Power BI dashboard and user evaluation.

A maximum of 15 important features was displayed in the global plots to keep them readable.

4.16 Important Feature Relationship Analysis

A limited number of feature relationship plots were created.

The selected features were:

- Tenure
- Monthly Charges

These features were selected because they were the most important continuous variables according to the grouped SHAP results.

The Tenure relationship plot showed that lower tenure values generally produced positive SHAP contributions, while longer tenure values usually produced negative SHAP contributions.

This means that newer customers were generally assigned greater churn risk, while long-term customers were generally assigned lower churn risk.

The Monthly Charges relationship plot showed that different charge levels influenced the churn probability differently.

Higher monthly charges frequently produced positive churn-risk contributions, although the exact contribution also depended on the customer's other features.

Only a limited number of relationship plots were created because the main purpose of this phase was to prepare structured outputs rather than generate unnecessary visualizations.

4.17 Local Customer-Level Explanations

Local SHAP explanations were created to explain individual customer predictions.

A waterfall plot was used for each selected example.

The waterfall plots show:

- The SHAP base probability
- Features increasing churn probability
- Features reducing churn probability
- The final customer churn probability

Positive contributions move the prediction toward higher churn risk.

Negative contributions move the prediction toward lower churn risk.

The local explanations demonstrate that two customers can receive different predictions because their features contribute in different directions and with different strengths.

4.18 Representative Customer Examples

Several representative customer cases were selected.

Example	Customer ID	Actual churn	Predicted churn	Churn probability
Highest churn probability	5178-LMXOP	1	1	0.859534
Lowest churn probability	0825-CPPQH	0	0	0.027568
Correctly predicted churn	5178-LMXOP	1	1	0.859534
Correctly predicted non-churn	0825-CPPQH	0	0	0.027568
False positive	5150-ITWWB	0	1	0.808563
False negative	3400-ESFU representative customer cases were selected.			

Example	Customer ID	Actual churn	Predicted churn	Churn probability
Highest churn probability	5178-LMXOP	1	1	0.859534
Lowest churn probability	0825-CPPQH	0	0	0.027568
Correctly predicted churn	5178-LMXOP	1	1	0.859534
Correctly predicted non-churn	0825-CPPQW	1	0	0.028512
Closest to threshold	1600-DILPE	0	1	0.270796

The customer with the highest churn probability was also the selected correctly predicted churn example.

The customer with the lowest churn probability was also the selected correctly predicted non-churn example.

This duplication occurred naturally because these customers satisfied both selection conditions.

The false-positive explanation shows why the model assigned high risk to a customer who did not actually churn.

The false-negative explanation shows why a real churn customer received a low model probability.

The customer closest to the threshold had a churn probability of approximately 0.2708, which was only around 0.0008 above the selected threshold of 0.27.

This example is useful for showing how a very small probability difference can change the final predicted label.

4.19 Creation of Customer Top-Factor Outputs

For every customer, the notebook identified:

- Top five positive SHAP factors
- Top five negative SHAP factors

The positive factors increased the churn probability.

The negative factors reduced the churn probability.

Each factor record contains:

- Customer ID
- Factor rank
- Factor type
- Original feature name
- Readable feature name
- Original customer value
- SHAP value
- Absolute SHAP value
- Contribution direction
- Readable explanation text
- Actual churn status
- Predicted churn status
- Churn probability
- Prediction threshold
- SHAP output scale

Example explanation text follows the format:

Contract = Month-to-month increased churn probability by approximately 0.1200.

The explanation text was created automatically using the original customer value and the probability-scale SHAP contribution.

These outputs will support customer-level Power BI charts, tables, and explanation cards.

4.20 Customer SHAP Summary Creation

A customer-level SHAP summary file was created with one record for every test customer.

The summary contains:

- Customer ID
- Actual churn status
- Predicted churn status
- Churn probability
- Prediction threshold
- SHAP base value
- Sum of SHAP contributions
- Reconstructed model output
- Number of positive contributions
- Number of negative contributions
- Number of neutral contributions
- Total absolute SHAP contribution
- Strongest positive factor
- Strongest negative factor
- Readable explanation text

The file contains 1,407 customer records.

A second master explanation file combined the customer SHAP summary with the original customer information from the Phase 4 prediction file.

This creates one main customer table containing:

- Original customer attributes
- Model prediction
- Churn probability
- SHAP summary
- Strongest positive factor
- Strongest negative factor

4.21 Creation of Power BI Long-Format Outputs

Two main long-format SHAP files were created for Power BI.

Processed-Feature Long Format

This file contains one row for every customer and every processed feature.

The total number of rows is:

$$1{,}407 \times 45 = 63{,}315 \text{ rows}$$

The file contains 20 columns.

The main fields include:

- Customer ID

- Processed feature name
- Original feature name
- Original feature value
- SHAP value
- Absolute SHAP value
- Contribution direction
- Global importance rank
- Customer contribution rank
- Actual churn
- Predicted churn
- Churn probability
- Prediction threshold
- SHAP output scale

Grouped Original-Feature Long Format

This file contains one row for every customer and every original feature.

The total number of rows is:

$$1{,}407 \times 19 = 26{,}733 \text{ rows}$$

The file contains 16 columns.

This is expected to be the main explanation table used in Power BI because it contains readable original features and customer values.

The long format will allow Power BI to:

- Filter explanations by customer
- Sort contributions by strength
- Display positive and negative factors
- Compare different customers
- Compare customer segments
- Display global and local explanation charts
- Support drill-through pages

4.22 Creation of Wide-Format Outputs

Wide-format SHAP files were also created.

The processed wide file contains one row per customer with separate columns for all 45 processed SHAP features.

The grouped wide file contains one row per customer with separate columns for all 19 grouped original SHAP features.

The wide files may support:

- Additional validation
- Customer-level data inspection
- Alternative Power BI calculations
- Future statistical analysis

The long-format files remain more suitable for most Power BI visualizations.

4.23 Final Phase 5 Validation

A complete final validation process was performed.

Sixteen validation checks were completed.

All sixteen checks passed.

The validation confirmed that:

- The correct Gradient Boosting model was loaded.
- Training data shape was correct.
- Testing data shape was correct.
- The exact feature order was preserved.
- Model probabilities matched Phase 4.
- SHAP customer count was correct.
- SHAP feature count was correct.
- All SHAP values were finite.
- Customer IDs remained aligned.
- The processed feature mapping was complete.
- Grouped SHAP values preserved the total contribution.
- SHAP additivity passed.
- Processed long-format files reloaded successfully.
- Grouped long-format files reloaded successfully.
- Customer summary files reloaded successfully.
- All required output files were present.

The final validation contained:

- 0 failed checks
- 0 missing customer explanations
- 0 missing SHAP feature values
- 0 infinite SHAP values
- 0 customer-ID alignment errors

4.24 Saving the Phase 5 Outputs

All Phase 5 outputs were stored inside the folder:

phase5_outputs

The following files were created:

Customer-Level Explanation Files

- customer_prediction_explanations.csv
- customer_shap_summary.csv
- customer_top_factors_combined.csv
- customer_top_negative_factors.csv
- customer_top_positive_factors.csv
- representative_customer_examples.csv

Global Explanation Files

- global_shap_importance_grouped.csv
- global_shap_importance_processed.csv

SHAP Data Files

- shap_values_grouped_long.csv
- shap_values_grouped_wide.csv
- shap_values_processed_long.csv
- shap_values_processed_wide.csv
- shap_feature_mapping.csv

Validation and Metadata Files

- shap_additivity_validation.csv
- phase5_validation_summary.csv
- shap_metadata.json

Global Visualization Files

- global_shap_bar_grouped.png
- global_shap_bar_processed.png
- shap_beeswarm_processed.png
- relationship_MonthlyCharges.png
- relationship_tenure.png

Local Waterfall Visualizations

- waterfall_highest_churn_probability_5178-LMXOP.png
- waterfall_lowest_churn_probability_0825-CPPQH.png
- waterfall_correctly_predicted_churn_5178-LMXOP.png
- waterfall_correctly_predicted_non_churn_0825-CPPQH.png
- waterfall_false_positive_5150-ITWWB.png
- waterfall_false_negative_3400-ESFUW.png
- waterfall_closest_to_selected_threshold_1600-DILPE.png

A final ZIP archive was also created:

Phase_5_SHAP_Outputs.zip

The archive contains 28 files and has an approximate size of 5.18 MB.

5. Purpose of the Main Output Files

shap_values_grouped_long.csv

This is the main Power BI-ready explanation table.

It contains one row for each customer and each original feature.

It includes the customer ID, original feature value, SHAP contribution, direction, model probability, prediction, and importance ranks.

shap_values_processed_long.csv

This file preserves the exact technical SHAP results for all 45 processed model features.

It will be useful for technical validation and detailed feature-level analysis.

global_shap_importance_grouped.csv

This file contains the global importance ranking of the 19 original customer features.

It will be used for readable global feature-importance visualizations in Power BI.

global_shap_importance_processed.csv

This file contains the importance ranking of the exact 45 processed model features.

It supports technical analysis of the model's encoded inputs.

customer_prediction_explanations.csv

This file combines original customer information, model predictions, churn probability, SHAP summaries, and strongest positive and negative explanations.

It can act as the main customer table in Power BI.

customer_shap_summary.csv

This file contains one explanation summary per customer.

It includes the base value, total SHAP contribution, reconstructed output, contribution counts, and strongest factors.

customer_top_positive_factors.csv

This file contains the five strongest risk-increasing factors for every customer where positive factors were available.

customer_top_negative_factors.csv

This file contains the five strongest risk-reducing factors for every customer where negative factors were available.

customer_top_factors_combined.csv

This file combines the positive and negative customer factors into one Power BI-friendly table.

shap_feature_mapping.csv

This file connects each of the 45 processed model features to its original customer feature and readable display name.

shap_additivity_validation.csv

This file confirms that the SHAP base value and customer contributions reconstruct the model output.

phase5_validation_summary.csv

This file records the final Phase 5 validation checks and confirms that all checks passed.

shap_metadata.json

This file records the SHAP configuration, including:

- Model name
- Model class
- Positive class
- Prediction threshold
- SHAP version
- Explainer type
- SHAP output scale
- Background sample size
- Dataset sizes
- Feature counts
- Grouping method
- Additivity tolerance
- Maximum additivity error
- Random state
- Output-file list

6. Tools and Libraries Used

The following tools and libraries were used during Phase 5:

- Google Colab

- Python
- Pandas
- NumPy
- Scikit-learn
- SHAP
- Matplotlib
- Joblib
- JSON
- ZIP file utilities
- Google Colab file utilities

SHAP was mainly used for:

- Tree-based model explanation
- Probability-scale feature contribution calculation
- Global feature importance
- Beeswarm visualization
- Local waterfall explanations

Pandas and NumPy were mainly used for:

- Loading the datasets
- Feature mapping
- SHAP output transformation
- Grouping encoded features
- Long-format table creation
- Customer-level factor ranking
- Validation

Matplotlib was used for:

- Grouped feature-importance plots
- Numerical feature relationship plots
- Saving the SHAP visualizations

7. Key Explainability Decisions

The main decisions made during Phase 5 were:

- Use the exact selected Gradient Boosting model from Phase 4.
- Preserve the exact order of the 45 processed features.
- Explain the positive class Churn = 1.
- Use SHAP TreeExplainer because the selected model is tree-based.
- Use a reproducible 100-record training background sample.
- Calculate SHAP values on the probability scale.
- Explain all 1,407 independent test customers.
- Validate that Phase 4 probabilities are reproduced.
- Validate SHAP additivity for every customer.
- Preserve all 45 processed-feature explanations.

- Group related one-hot encoded contributions into 19 original features.
- Validate that grouping preserves the total contribution.
- Connect SHAP results to customer IDs.
- Use original customer values instead of standardized values for readable outputs.
- Create global and local explanation outputs.
- Create the five strongest positive and negative factors per customer.
- Create long-format tables for Power BI.
- Create wide-format tables for validation and alternative analysis.
- Avoid excessive visualizations.
- Save the SHAP scale and configuration in metadata.
- Treat SHAP values as explanations of model behaviour, not proof of causal relationships.

8. Key Outcomes

At the end of Phase 5:

- The selected Gradient Boosting model was loaded successfully.
- The Phase 4 probabilities were reproduced correctly.
- All 1,407 test customers received SHAP explanations.
- Each customer received 45 processed-feature contributions.
- SHAP values were calculated on the probability scale.
- No missing SHAP values were found.
- No infinite SHAP values were found.
- Additivity passed for all 1,407 customers.
- The 45 processed features were mapped successfully.
- The processed features were grouped into 19 original features.
- Grouped SHAP values preserved the total contribution.
- Tenure was the most important original feature.
- Contract was the second most important original feature.
- Customer-level positive and negative factors were created.
- Representative correct and incorrect predictions were explained.
- Global SHAP visualizations were created.
- Local waterfall explanations were created.
- Power BI-ready long-format outputs were created.
- Customer-level summary outputs were created.
- All sixteen final validation checks passed.
- Twenty-eight output files were included in the final ZIP archive.
- The SHAP explanation outputs are now ready for Power BI.

9. Connection to the Power BI Dashboard

Phase 5 created the explanation layer required for the interactive Power BI dashboard.

The Power BI dashboard will combine:

- Original customer information
- Actual churn status
- Predicted churn status

- Churn probability
- Prediction threshold
- Global SHAP importance
- Customer-level SHAP contributions
- Top positive churn factors
- Top negative churn factors
- Contribution direction
- Original customer feature values

The main Power BI tables are expected to include:

- customer_prediction_explanations.csv
- shap_values_grouped_long.csv
- customer_top_factors_combined.csv
- global_shap_importance_grouped.csv
- Phase 4 model-performance outputs

The tables can be connected through:

customerID

The grouped SHAP table will support:

- Customer filtering
- Positive and negative contribution charts
- Individual customer drill-through
- Contribution-strength ranking
- Comparison between customer groups
- Explanation of high-risk customers
- Examination of false-positive and false-negative cases

The global importance table will support a dashboard view showing which customer features influence the model most strongly overall.

The customer explanation table will support cards and summaries showing:

- Churn probability
- Predicted churn label
- Actual churn label
- Strongest factor increasing risk
- Strongest factor reducing risk

10. Next Steps

The next phase of the thesis will focus on developing the interactive Power BI visual analytics system.

The planned tasks include:

1. Select the final Phase 4 and Phase 5 files required for Power BI.
2. Import the customer prediction and SHAP files into Power BI.
3. Verify data types and customer-ID relationships.
4. Build the Power BI data model.
5. Create an Executive Overview page.
6. Create a Customer Risk Analysis page.
7. Create a Global Model Explanation page.
8. Create an Individual Customer Explanation page.
9. Create a Customer Segment Comparison page.
10. Create a Model Performance Summary page.
11. Display churn probabilities and predicted churn labels.
12. Display global grouped SHAP importance.
13. Display positive and negative customer contributions.
14. Add customer and segment filters.
15. Add drill-through functionality.
16. Add tooltips and readable explanation text.
17. Check that contribution direction is displayed correctly.
18. Create a static explanation baseline.
19. Compare the static and interactive system designs.
20. Prepare the Phase 6 progress documentation.

11. Summary

Phase 5 focused on applying Explainable Artificial Intelligence to the selected customer churn model.

The trained Gradient Boosting model from Phase 4 was loaded together with the processed datasets, original customer records, saved predictions, feature names, and model metadata.

The model and data were validated before generating explanations. The saved churn probabilities were reproduced successfully, and the customer IDs and feature order remained correctly aligned.

SHAP TreeExplainer was used with an interventional background dataset containing 100 training customers. The explanations were generated directly on the probability scale for the positive churn class.

SHAP values were calculated for all 1,407 test customers and all 45 processed features. The explanations contained no missing or infinite values.

Additivity validation confirmed that the SHAP base value and feature contributions reconstructed the model probability for every customer.

The 45 processed features were also grouped into 19 original customer features. This created more understandable explanations for business users and future Power BI integration.

The global results showed that Tenure, Contract, Internet Service, Online Security, Technical Support, Payment Method, and Monthly Charges were among the most important features used by the model.

Customer-level explanation outputs were created to identify the strongest factors increasing and reducing churn risk. Representative examples were also generated for high-risk, low-risk, correctly predicted, false-positive, false-negative, and threshold-level customers.

Processed and grouped SHAP outputs were saved in both long and wide formats. The grouped long-format file contains 26,733 rows and will be the main explanation table used during Power BI development.

All final validation checks passed successfully.

The prediction and explanation components of the proposed visual analytics system are now complete. The project is ready to continue with Phase 6, which will focus on designing and implementing the interactive Power BI dashboard.